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Mathematics

ENGLISH

10 Standard

Part –2

Version 1.1

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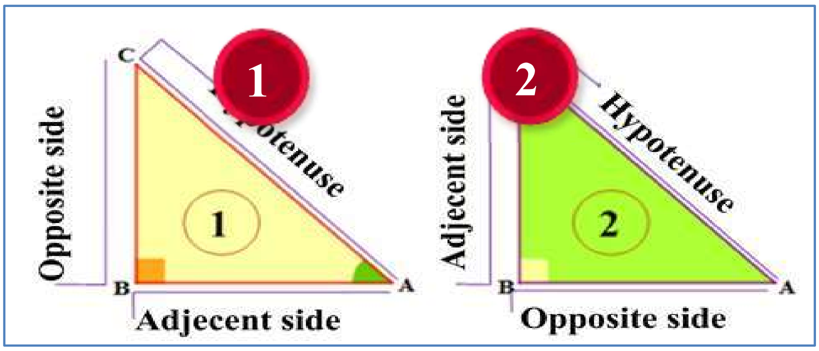
8 ITRODUCTION TO TRIGONOMETRY

Trigonometry is the study of relationships between the sides and angles of a triangle.

11.2 Trigonometric Ratios

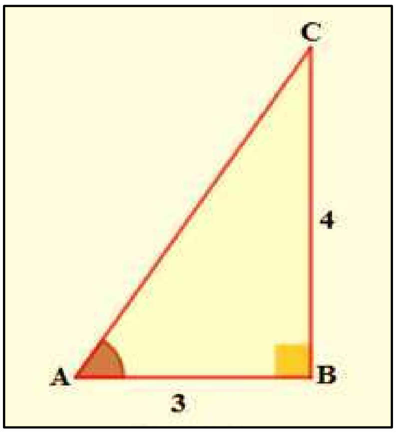
To know the trigonometric ratio we have to consider right angle triangle.



Trigonometric ratios		1	2
SinA	$\frac{\text{Opposite}}{\text{Hypotenuse}}$	$\frac{BC}{AC}$	$\frac{AB}{AC}$
CosA	$\frac{\text{Adjecent}}{\text{Hypotenuse}}$	$\frac{AB}{AC}$	$\frac{BC}{AB}$
Tan A	$\frac{\text{Opposite}}{\text{Adjecent}}$	$\frac{BC}{AB}$	$\frac{AB}{BC}$
CosecA	$\frac{\text{Hypotenuse}}{\text{Opposite}}$	$\frac{AC}{BC}$	$\frac{AC}{AB}$
SecA	$\frac{\text{Hypotenuse}}{\text{Adjecent}}$	$\frac{AC}{AB}$	$\frac{AC}{BC}$
CotA	$\frac{\text{Adjecent}}{\text{Opposite}}$	$\frac{AB}{BC}$	$\frac{BC}{AB}$

There are six trigonometric ratios:

Example 1: tan A = $\frac{4}{3}$ find the other trigonometric ratios of the angle A



In ΔABC, $\angle ABC = 90^0$

$\therefore$ By Pythagoras theorem,

$AC^2 = AB^2 + BC^2$

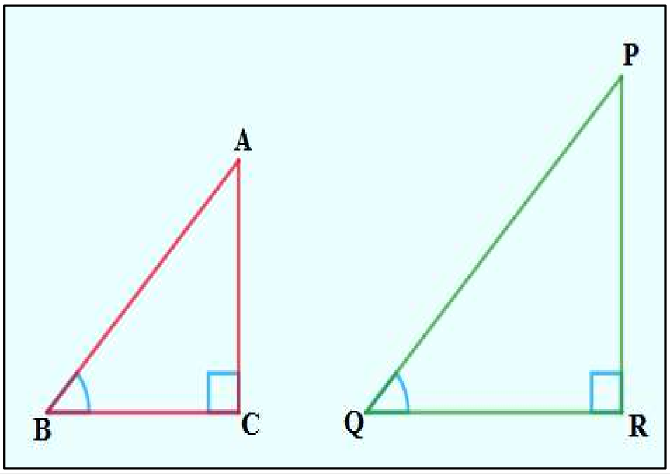
$\Rightarrow AC^2 = 4^2 + 3^3 = 16 + 9 = 25$

$\Rightarrow AC = 5$

$\text{Sin}A = \frac{BC}{AC} = \frac{4}{5}$; $\text{Cos } A = \frac{AB}{AC} = \frac{3}{5}$; $\text{Tan}A = \frac{BC}{AB} = \frac{4}{3}$

$\text{Cosec}A = \frac{AC}{BC} = \frac{5}{4}$; $\text{Sec}A = \frac{AC}{AB} = \frac{5}{3}$; $\text{Cot}A = \frac{AB}{BC} = \frac{3}{4}$

Example 2 :If B and Q are acute angles such that sin B = sin Q, then prove that $\angle B = \angle Q$.



Sin B = SinQ

$\Rightarrow \frac{AC}{AB} = \frac{PR}{PQ}$

$\Rightarrow \frac{AC}{PR} = \frac{AB}{PQ} = k$ ----- (1)

$BC = \sqrt{AB^2 - AC^2}$ [By Pythagoras theorem]

$\Rightarrow \sqrt{k^2PQ^2 - k^2PR^2}$

$\Rightarrow k.\sqrt{PQ^2 - PR^2}$ [from (1)]

$QR = \sqrt{PQ^2 - PR^2}$

$\Rightarrow \frac{BC}{QR} = \frac{k.\sqrt{PQ^2 - PR^2}}{\sqrt{PQ^2 - PR^2}} = k$ ------(2)

From (1) and (2),

$\frac{AC}{PR} = \frac{AB}{PQ} = \frac{BC}{QR}$

$\Rightarrow \Delta ABC \sim \Delta PQR$

$\therefore$ **∠B = ∠Q**

Example 3 : Consider ΔACB , right-angled at C, in which AB = 29 units, BC = 21 units and ∠ABC = θ (see Fig. 8.10). Determine the values of
$\cos^2\theta + \sin^2\theta$ (ii) $\cos^2\theta - \sin^2\theta$

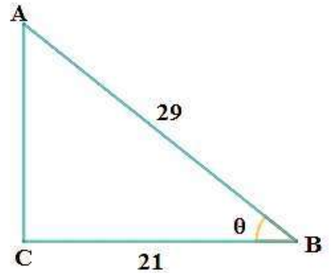


Fig. 8.10

In right angle triangle ACB, $\angle ACB = 90^0$

Therefore $AC = \sqrt{AB^2 - BC^2}$

$\Rightarrow AC = \sqrt{29^2 - 21^2}$

$\Rightarrow AC = \sqrt{841 - 441} = \sqrt{400} = 20$

(i) $\cos^2\theta + \sin^2\theta = \frac{21^2}{29^2} + \frac{20^2}{29^2} = \frac{441+400}{841} = \frac{841}{841} = 1$

(ii) $\cos^2\theta - \sin^2\theta = \frac{21^2}{29^2} - \frac{20^2}{29^2} = \frac{441-400}{841} = \frac{41}{841} = 1$

Example 4 : In a right triangle ABC, right-angled at B, if tan A = 1, then verify that 2sin A cos A = 1

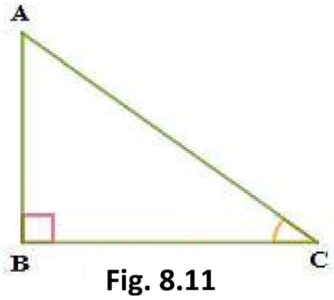


Fig. 8.11

In right angle triangle ACB ,

$\tan A = 1 \Rightarrow \frac{AB}{BC} = 1$

$\Rightarrow AB = BC$

$AC^2 = AB^2 + BC^2$ [By Pythagoras theorem]

$\Rightarrow AC^2 = 2AB^2$ -------- (1)

Now, $2\sin A \cos A = 2.\frac{AB}{AC}.\frac{BC}{AC}$

$= 2.\frac{AB^2}{AC^2}$

$= 2.\frac{AB^2}{2AB^2} = 1$

Example 5 : In ΔOPQ right-angled at P, OP = 7 cm and OQ – PQ = 1 cm (see Fig. 11.12). Determine the values of sin Q and cos Q.

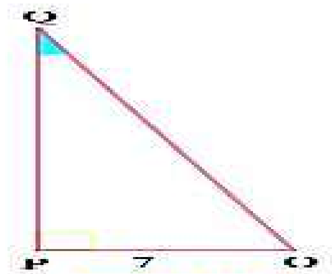


Fig. 8.12

In ΔOPQ, $OQ^2 = PQ^2 + OP^2$ [By Pythagoras theorem]

$\Rightarrow (1 + PQ)^2 = PQ^2 + 7^2$

$\Rightarrow 1 + PQ^2 + 2PQ = PQ^2 + 49$

$\Rightarrow 1 + 2PQ = 49$

$\Rightarrow 2PQ = 49 - 1 = 48$

$\Rightarrow PQ = 24cm$

$\Rightarrow OQ = 1 + PQ = 1 + 24$

$\Rightarrow OQ = 25$

$\therefore$ **$\sin Q = \frac{7}{25}$ and $\cos Q = \frac{24}{25}$**

Inverse of trigonometric values

$\frac{1}{\text{SinA}}$	$\frac{\text{Hypotenuse}}{\text{Opposite}}$	**CosecA**
$\frac{1}{\text{CosA}}$	$\frac{\text{Hypotenuse}}{\text{Adjecent}}$	**SecA**
$\frac{1}{\text{Tan A}}$	$\frac{\text{Adjecent}}{\text{Opposite}}$	**CotA**

Inverse of trigonometric values		
$\frac{1}{\text{CosecA}}$	$\frac{\text{Opposite}}{\text{Hypotenuse}}$	**SinA**
$\frac{1}{\text{SecA}}$	$\frac{\text{Adjecent}}{\text{Hypotenuse}}$	**SecA**
$\frac{1}{\text{CotA}}$	$\frac{\text{Opposite}}{\text{Adjecent}}$	**CotA**